

Mahdi Shakouri

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Summary

- **Languages and Libraries:** Python, Java, C/C++, MATLAB, PyTorch, TensorFlow, OpenCV, LaTeX, SQL, URDF
- **Tools and Platforms:** Git/GitHub, Linux, ROS, Gym-gazebo, Altium, Qt, ImageJ, Webots, MQTT, Isaac Lab
- **Hardware Skills:** PCB Design, CAD, Rapid Prototyping, 3D-Printing, Circuit Design, Soldering, SolidWorks

Education

Bachelor of Applied Science, Engineering Physics
University of British Columbia

Sep 2021 – May 2026

- **Core Courses:** Calculus III and IV, Applied Linear Algebra, PDEs, Applied PDEs, Probability, Stochastic Processes, Complex Analysis, Electromagnetic Theory, Classical Mechanics, Signals & Systems, Classical Control Systems, Software Construction, Neural Networks, Deep Learning, Computational Physics, Quantum Mechanics, Cybersecurity
- **Awards:** Dean's Honor List – Guenter Felix Sanders Scholarship, BC Excellence Entry Scholarship – CGPA: 83%

Experiences

Algorithms Engineer Intern
Kardium Inc.

May – Dec 2024 | May – Aug 2025
Vancouver, BC

- Worked on improving the navigational capabilities of the Globe™ Catheter, delivering Pulse-Field ablation to cardiac tissue for atrial fibrillation therapies using generated multi-planar RF electric fields.
- Used the SAM2 deep learning model for device localization in multi-planar fluoroscopic images, significantly reducing annotation time of an hour-long procedure from 8 hours manually to 6 mins quasi-automatically.
- Filtered and used the SAM2 results to train YOLOv9 for fully automated segmentation in 2D fluoroscopy videos.
- Developed a localization algorithm to pinpoint the Globe™ location with sub-millimeter accuracy with PCA in 3D.
- Improved the accuracy of maps by 8% by implementing a graph-based algorithm to identify and remove outliers in cardiac electrograms, enhancing the precision of cardiac mapping. Reduced false positives in LAT maps by 42%.
- Compiled a dataset of tissue contact and temperature measurements from 250+ clinical cases using short-term reliable multi-variate data, supporting the development of machine learning models for real-time contact assessment.
- Quantified accuracy of measurements of device proximity to tissue using thermal dispersion by analyzing 1TB of clinical data and cross-validating with known bench-top measurements using multiple regression models.

Undergraduate Teaching Assistant
UBC Faculty of Physics and Astronomy

Sep 2023 – Present
Vancouver, BC

- Lead tutorials & office hours for Neural Networks ([ENPH 353](#)), Intro Physics ([PHYS 157](#)), and E&M ([158](#)) courses.

Undergraduate Research Assistant
McKeown Lab, Center for Brain Health

Jan – May 2024
Vancouver, BC

- Augmented video feed data by manipulating video quality for testing a vision-based assessment pipeline for Parkinson's disease using computer-vision methods.
- Conducted literature review to contribute to a manuscript on video quality and remote Parkinson's assessment.

Tissue Engineering Intern
ICORD and Center for Blood Research

Jan – Aug 2023
Vancouver, BC

- Developed imaging software with blob detection, modified Laplacian filters, SIFT, and directional analysis using MATLAB and Python to quantify ganglion characteristics, outgrowth and alignment in in-vitro growth samples.
- Designed, prototyped, and fabricated multiple iterations of 3D-printed microfluidic chips modelling the intervertebral discs for spinal cord injury experiments with a focus on neuronal regeneration.
- Manufactured microfluidic chips and completed laboratory protocol analysis to determine efficacy of hydrogels.
- Conducted ICC protocol and imaged cultured samples using Confocal and Fluorescence microscopy.

Curriculum Designer & Instructor
UBC Geering Up

Apr – Aug 2022
Vancouver, BC

- Developed and instructed week-long curriculum for Intro to Object Oriented Programming for Gr. 10-12 summer camps with mini-projects and interactive challenges with Python for high-school students.

Technical Projects

Sim2Real Multi-Agent RL Driving

Sep 2025 - Current

Capstone Sponsored by Wayve.ai

([GitHub](#))

- Creating an overhead visual tracking system that can extract full-state positions of all cars in real-time.
- Training and deploying a PPO RL policy for a single vehicle to navigate a course safely in Isaac Lab and reality.
- Implementing heterogeneous actuation & sensing limits across agents with digital twin simulation for offline training.
- Interfaced a micro-RC car to external control via a ESP32 microcontroller using MQTT for wireless communication.

Conditional PixelCNN for Generation

Mar – Apr 2025

Python, PyTorch, Deep Learning

([Project Description](#) | [GitHub](#))

- Implemented a conditional PixelCNN in PyTorch to generate images conditioned on 4 class labels trained with NLL.
- Achieved a 76.64% accuracy and an FID score of 30.80 on the CPEN455 test dataset in classification and generation.

RF Interference Monitoring System for the Dominion Observatory

Sep – Apr 2025

Signal Processing, Impedance Matching, Statistical Mechanics, C++

([Project Report](#))

- Designed and deployed a broadband radio-frequency interference monitoring instrument to detect and report Ultra-High Frequency (UHF) noise bursts from maintenance vehicles and electronics on the site access road in the 400-800 MHz range using a combination of monopole and Log-periodic antennae and Power Spectral Density measurements.
- Implemented a user interface on a Raspberry Pi and an optical fibre communication between the device and off-site users to access and extract interference reports with pre- and post-processed data generated by the on-site device.

Simulated Self-Driving Vehicle

Sep – Dec 2024

OpenCV, Python, TensorFlow, ROS, Linux, Machine Learning

([Project Description](#) | [GitHub](#))

- Developing a computer vision algorithm capable of navigating an autonomous vehicle in a simulated world.
- Implemented a convolution neural network for recognizing objects and signs within the world to make decisions.
- Created and trained an imitation learning model by collecting labelled data from manually driven trials in the world.
- Trained a reinforcement learning model to maneuver the vehicle to target locations in the world to collect points.

Autonomous Race Robot

May – Aug 2023

SolidWorks, Altium, C/C++, Circuit Design, MCU, PCB Layout

([Project Description](#) | [GitHub](#))

- Designed, prototyped, and programmed a race robot from scratch with waterjet, laser-cut, and 3D-printed parts.
- Designed a state machine architecture firmware for automating the robot, interfaced using a 32-bit microcontroller.

Multi-client Twitter Server

Nov – Dec 2022

Threading, Java, Object-Oriented Programming

([Project Description](#))

- Developed a multi-client Twitter server in Java using sockets, threading, and message queues to handle concurrent client queries. Learned about multi-threading, concurrency, and network programming.

Presentations and Posters

- Shakouri M., Akhavizadegan M., “From Observations to Conclusions: A Systemic Approach”, Az-Zahraa Systems Thinking Symposium, 2024, Oral Presentation, Vancouver, B.C., Canada
- Shani B., Shakouri M., Aliabadi N., “Distribution of Covid-19 Outbreaks in Relation to Avg. Family Income in Communities in B.C.”, 2022, STEM Fellowship Big Data Challenge, Poster, Vancouver, B.C., Canada
- Gibbs W., Shakouri M., Ghasaei S., Garavelli F., “The Electro-Mechanical Workings of a 3D Printer”, Engineering Physics 2nd Year Presentations, 2023, Oral Presentation, Vancouver, B.C., Canada

Scholarships and Awards

- UBC Affiliate Scholarships | \$1150 | 2024 | UBC
- Outstanding Junior Referee | 2022 | Volleyball BC
- Guenter Felix Sanders Scholarship | \$1000 | 2021 | University Hill Secondary
- BC Excellence Entry Scholarship | \$5000 | 2021 | BC Ministry of Education
- BC Achievement Scholarship | \$1250 | 2021 | BC Ministry of Education
- Loran Scholars Semi-Finalist | 2021 | Loran Scholars

Volunteer & Community Endeavours

President | VP Events

2023/24 | 2024/25

Thaqalayn Muslim Association of UBC (TMA UBC)

UBC Vancouver

- Led a team of students to organize cultural, social, and career events for the UBC Shia community.
- Strived to foster a safe and welcoming community and environment where students from all backgrounds and ethnicities could feel welcomed and gather around their shared values
- Organized small- and large-scale networking conferences including our annual gala with over 500 attendees.
- Lead the club to hosting its first ever BC Shia Conference events with over 400 attendees and a budget of \$12000.
- Established meaningful relationships with other faith-based student clubs and organized interfaith and multifaitth collaborations and events to bring the broader student community at UBC together & unite over our shared values.

Weekend School Teacher

Sep 2021 – Jun 2024

Az-Zahraa Academy

Richmond, BC

- Bringing youth closer to their culture and heritage with engaging activities and teaching them how to read Arabic.
- Spend hours planning, brainstorming, and implementing educational activities and group projects, and practicing poem recitals with students, aiming to engage my students intellectually.

Sports Representative and Director

Sep 2023 – Aug 2025

FIZZ (UBC Engineering Physics Student Council)

Vancouver, BC

- Organized and coordinated the sporting activities of the Engineering Physics Student Council including participation in intramural sports, recreational activities, and competitions like Storm the Wall.
- Encouraged the participation of ENPH students in community building and physical activities through social media.

Student Mentor

Jan 2024 – Current

Vancouver Nour Danesh Academy (VANDA Academy)

North Vancouver, BC

- Helping Persian immigrant youth to maintain their identity and thrive in the Canadian culture and education system.
- Sharing opportunities and experiences with youth, supporting their career path development and their pursuit of post-secondary education through academic planning meetings, profile writing workshops, and 1-on-1 consultations.

Volleyball Coach

Sep 2020 – Jan 2024

University Hill Secondary | Volleyball BC

Vancouver, BC

- Fostered an exciting and productive environment for youth athletes to grow, both as athletes and caring individuals.
- Enriched and enhanced young athletes' teamwork and volleyball skills through designing practice drills and workouts.
- Nourished and encouraged youth players grow together, both as athletes and caring individuals in their lives.

Certified Referee

Sep 2020 – Current

Volleyball Canada | Volleyball BC

Lower Mainland, BC

- Officiate local, regional, and national youth events as a certified official, upholding Volleyball Canada standards.

Summer Camps Volunteer

May 2019 – Aug 2021

Vancouver Aquarium

Vancouver, BC

- As a Volunteer Camp Leader, assisted staff in keeping campers engaged, educated campers about our oceans and ecosystem, and brought nature awareness through leading related activities and games.
- Learned to effectively collaborate with staff members and appreciate the unique characters of every child.
- Created an entertaining and educational atmosphere for over 100 children while keeping them physically and intellectually active.

Volunteer Student Researcher

May 2020 – Dec 2020

BC Cancer Research Institute

Vancouver, BC

- Created a searchable and customized database of the selected samples on Excel for BC Cancer scientists to help their research on AI-aided prognostics. Also visualized DICOM images with Python & Octave for quality control.